

Advanced Python

Day 9

Magic Methods: `__str__`, `__repr__`, `__len__`, `__add__`

Making Objects Smart

Goal: Make your custom objects behave in a smart, readable, and native way.

The Product Label Analogy

Imagine buying a laptop. The label reads:

- **Product:** Laptop
- **Price:** ₹50,000
- **Brand:** Dell

This label clearly tells you what the object is.

But imagine the shop instead prints:

`Object#4929384 MemoryAddress`

That would be completely useless! This is exactly what happens in Python without using `__str__()`.



 Copilot+PC

Python Without Magic Methods

The Code

```
class Car:
    def __init__(self, brand,
price):
        self.brand = brand
        self.price = price

c1 = Car("Tesla", 80000)
print(c1)
```

The Output

```
<__main__.Car object at
0x000002...>
```

This is like a product without a proper label. It only shows the system memory address, which is **not useful** for humans trying to read the data.

Human-Friendly with `__str__()`

The Code

```
class Car:
    def __init__(self, brand,
price):
        self.brand = brand
        self.price = price

    def __str__(self):
        return f"Car: {self.brand},
₹{self.price}"

c1 = Car("Tesla", 80000)
print(c1)
```

The Output

```
Car: Tesla, ₹80000
```

Now we give our object a proper label! The `__str__()` method defines exactly how the object should look when printed to the console.

Real-Life Meaning of `__str__()`



Product Label

Clearly describes the item's details, brand, and exact price to the consumer.



ID Card

Provides an immediate summary of an individual's identity and key information.



Name Tag

A simple, human-readable identifier used to make interactions easier.

Container Behavior: `__len__()`

Real-Life Example

Imagine a physical basket holding several items.

Python naturally wants to ask: *"How many items are inside?"*

By defining the `__len__()` magic method, you instruct Python on how to calculate size. Your object now behaves like a standard Python list!

The Code

```
class Basket:
    def __init__(self, items):
        self.items = items

    def __len__(self):
        return len(self.items)

b = Basket(["apple", "mango"])
print(len(b)) # Output: 2
```


Mathematical Behavior: `__add__()`

Real-Life Example

Combining two physical money wallets into one total sum.

The `__add__()` magic method allows you to use the standard `+` mathematical operator directly on your objects.

Your custom objects now behave seamlessly like numbers.

The Code

```
class Wallet:
    def __init__(self, money):
        self.money = money

    def __add__(self, other):
        return self.money +
        other.money

w1 = Wallet(100)
w2 = Wallet(200)
print(w1 + w2) # Output: 300
```


Developer Insight

-  **Seamless Integration:** Magic methods allow your custom objects to act seamlessly like native Python objects (numbers, lists, strings).
-  **Under the Hood:** Built-in functions rely entirely on these methods. When you call `len(list)`, it triggers `__len__()` internally.
-  **Automatic Formatting:** Statements like `print(object)` automatically invoke the `__str__()` method.
-  **Intuitive Operators:** Even basic math operations like `a + b` are translated to `a.__add__(b)` by Python behind the scenes.

This is exactly why Python libraries feel incredibly natural to use.

Practice Tasks

Task 1 & 2

- 1. Product System:** Create class `Product` with name and price. Use `__str__()` to neatly display the info.
- 2. Library Book:** Create class `Book` with title and author. Print the object using `__str__()`.

Task 3 & 4

- 3. Shopping Cart:** Create class `Cart` holding an items list. Implement `__len__()` to return the count.
- 4. Bank Wallet:** Create class `Wallet` holding money. Implement `__add__()` to combine balances.

Day 9 Summary

What You Learned:

- ✓ **Magic Methods:** How they empower classes.
- ✓ **`__str__()`:** Creating human-friendly labels.
- ✓ **`__len__()`:** Defining container behavior.
- ✓ **`__add__()`:** Enabling native math operations.

🔥 Next Up: Day 10

Static Methods & Class Methods (Real company system example)



Day 9 Advanced Done!

Ready for Day 10?

Image Sources



https://dellstatic.luroconnect.com/media/catalog/product/cache/74ae05ef3745aec30d7f5a287debd7f5/x/p/xps-13-9350-laptop-sl-copilot-pc-mg_oxn9350110001rino_1.png

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